(this table was in original submission)

Table 6: Accuracy (%) intervening on the set of 28 `addition` neurons $\mathcal{N}_{\text{add}}$. **Clean**: Llama-3.1-8B accuracy on this task. **Only $\mathcal{N}_{\text{add}}$**: accuracy when all other neuron activations at the layer 18 MLP are set to 0, except for neurons in $\mathcal{N}_{\text{add}}$. Despite these neurons making up 0.2% of layer 18 neurons, accuracy remains high. **Zeroed**: accuracy drops significantly when these neurons' activations are set to zero. **Flipped**: reversing the sign of these neurons' activations is destructive, likely because they represent periodic functions that oscillate between negative and positive values across inputs.

Task	Number Range	Clean	Only $\mathcal{N}_{\text{add}}$	Zeroed	Flipped	*Chance*
Weekdays	1→7	91.84	89.80	40.82	20.41	*14.29*
	8→14	68.37	58.16	28.57	17.35	*14.29*
Months	1→12	81.25	77.78	36.81	14.58	*8.33*
	13→24	64.93	70.83	23.61	16.32	*8.33*
Hours	1→24	93.06	80.56	50.69	16.32	*4.17*
	24→48	69.18	51.65	29.77	11.63	*4.17*
Addition	$a, b \in [1..100]$	94.60	86.36	24.11	2.97	*0.50*

(1) Ablations for 16 additional templates ⬇️

Table 7: Neuron intervention results for the `addition` task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	Input Format	Offset Format	Clean	Only ↑	Zeroed ↓	Flipped ↓	Random ↑
(default prompt)	1, 2...	1, 2...	94.33%	86.11%	23.80%	2.92%	94.47% ± 0.17
The sum of {input} and {offset} equals	1, 2...	1, 2...	100.00%	98.87%	42.74%	5.09%	100.0% ± 0.00
Math question: {input} plus {offset} equals	1, 2...	1, 2...	96.76%	76.98%	29.30%	4.22%	96.66% ± 0.32
Q: What is the sum of {input} and {offset}? A:	1, 2...	1, 2...	99.70%	97.79%	43.93%	5.41%	99.68% ± 0.06
Q: If I have {input} apples and buy {offset} more, how many apples do I have? Total apples:	1, 2...	1, 2...	75.67%	40.63%	12.32%	1.43%	74.92% ± 2.47

Table 8: Neuron intervention results for the `weekdays` task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	**Input Format**	**Offset Format**	**Clean**	**Only** ↑	**Zeroed** ↓	**Flipped** ↓	**Random** ↑
(default prompt)	Monday, Tuesday...	one, two...	91.84%	89.80%	40.82%	20.41%	92.45% ± 1.31
Let's do some weekday math. {offset} days after {input} is what day? Answer:	Monday, Tuesday...	one, two...	73.47%	73.47%	30.61%	16.33%	73.47% ± 1.83
Question: Today is {input}. In {offset} days, what day will it be? Answer: It will be	Monday, Tuesday...	1, 2...	97.96%	85.71%	44.90%	24.49%	95.31% ± 1.84
If today is {input}, and I have a deadline in {offset} days, what day is my deadline? My deadline is on	Mon, Tues...	one, two...	83.67%	71.43%	40.82%	20.41%	86.53% ± 1.63
Let's do some calendar math. Today is {input}. What day will it be in {offset} days? Answer:	Mon, Tues...	1, 2...	85.71%	83.67%	38.78%	30.61%	83.47% ± 1.70

Table 9: Neuron intervention results for the `months` task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	**Input Format**	**Offset Format**	**Clean**	**Only** $\uparrow$	**Zeroed** $\downarrow$	**Flipped** $\downarrow$	**Random** $\uparrow$
(default prompt)	January, February...	one, two...	81.25%	76.39%	36.81%	14.58%	81.32% $\pm$ 1.00
Let's do some calendar math. If the current month is {input}, then in {offset} months, what month will it be? Answer: It will be	January, February...	one, two...	97.22%	96.53%	59.03%	22.22%	96.74% $\pm$ 0.82
Question: I have a big presentation in {offset} months. If it is currently {input}, then what month is my presentation? Answer: My presentation is in	January, February...	1, 2...	70.83%	64.58%	46.53%	25.00%	70.97% $\pm$ 0.52
Q: A pregnant woman will give birth in {offset} months. If it is currently {input}, what month will she give birth? Answer: She will give birth in	Jan, Feb...	one, two...	91.67%	89.58%	49.31%	17.36%	91.88% $\pm$ 1.28
Let's figure out some scheduling. I just got my hair cut, and it is currently {input}. If I have to book another appointment in {offset} months, what month will it be? Answer: My appointment is in	Jan, Feb...	1, 2...	81.25%	78.47%	53.47%	34.03%	80.90% $\pm$ 1.33

Table 10: Neuron intervention results for the `hours` task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	**Input Format**	**Offset Format**	**Clean**	**Only** ↑	**Zeroed** ↓	**Flipped** ↓	**Random** ↑
(default prompt)	00, 01...	one, two...	97.57%	84.55%	50.87%	16.15%	97.24% ± 0.28
Let's do some clock math. If it is {input}:00 right now in 24-hour time, then what time will it be in {offset} hours? Answer:	00, 01...	one, two...	90.45%	68.40%	47.40%	13.37%	90.31% ± 1.30
When I was in the military, I learned 24-hour time. So if right now it is {input}:00, then in {offset} hours, it will be	00, 01...	one, two...	91.32%	83.68%	50.52%	18.06%	91.15% ± 0.28
Q: If it is {input} o' clock in military time, what time will it be in {offset} hours? A: In 24-hour time, it will be	00, 01...	one, two...	81.08%	73.26%	47.57%	23.26%	81.23% ± 0.31
I need to work out when to get to the airport. Q: It's {input}:00 right now in 24-hour time. My flight is in {offset} hours. What time is my flight? Answer: My flight will be at	00, 01...	1, 2...	86.46%	64.58%	34.20%	12.67%	86.22% ± 0.59

(2) Sweep over neuron thresholds ⬇️

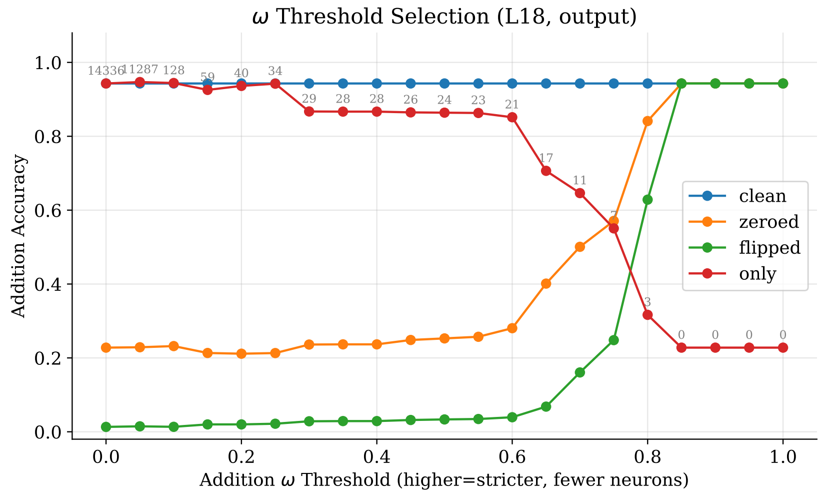

Figure 58: Sweeping over ω for the `addition` task. While we choose $\omega > 0.4$, keeping more neurons ($\omega > 0.2$) allows for cleaner results with still only 40 addition neurons. Gray text indicates the number of neurons included for this threshold. We include analysis on the additional neurons not in $\mathcal{N}_{add}$ in App. H.6.

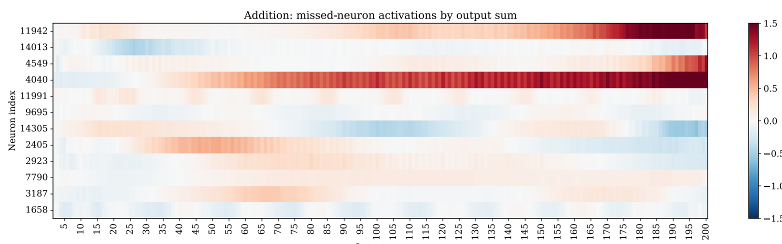

Figure 59: Activations for twelve extra neurons that are included when $\omega > 0.2$ for the `addition` task. These neurons also show wavelike patterns when activations are averaged across output sums.

H.6 Extra Addition Neurons

After sweeping over possible thresholds for ω for the `addition` task in Figure 58, we notice that cleaner ablation results can be achieved with a more lax threshold, including layer 18 MLP neurons with $\omega > 0.2$.

We plot the activation patterns for the additional twelve neurons (40 - 28) not yet analyzed in this paper in Figure 59, observing that they also have periodic patterns, with activations oscillating along output sums. In fact,

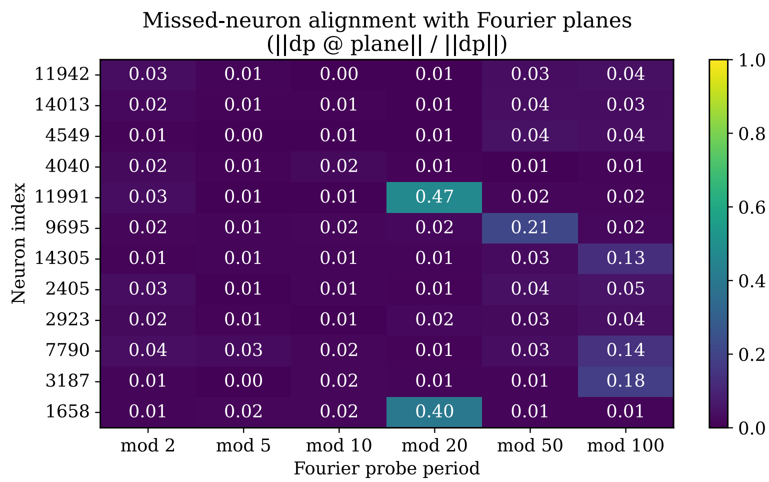

Figure 60: Alignment of twelve extra neurons from Figure 59 that are included when $\omega > 0.2$ for the `addition` task. Two neurons (11991, 1658) appear to be period 20 neurons that could be included in $\mathcal{N}_{add}$. Neuron 9695 appears aligned with $T = 50$, and neurons 14305, 7790, and 3187 are slightly aligned with $T = 100$. For other neurons, we see low alignment—we hypothesize that this is because these neurons write to larger periods. We can directly observe this in Figure 59: e.g., neuron 2405 is positive at 50 and negative at 200, suggesting that it has a period of $T = 300$, and neurons 11942 and 4040 appear to gradually activate for a large range of sums. Also see Figure 43.